PROJECT PHASE I REPORT

On

Implementation and Evaluation of Wind Profile Estimation Algorithms for CUSAT ST Radar

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In partial fulfilment of the requirements for the award of the degree of B.Tech in Electronics and Communication Engineering



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CERTIFICATE

Certified that the project phase - 1 report entitled ***"Implementation and Evaluation of Wind Profile Estimation Algorithms for CUSAT ST Radar"*** *is a bonafide work of* ***Naveen C Lal (20323079), Roopesh O R (20323085) and Sneha S (20323094)*** *towards the partial fulfilment of requirement for the award of the degree of B.Tech in Electronics and Communication of Cochin University of Science and Technology, Kochi-682022.*

Project Guide **Head of the Division**

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Abstract

Accurate wind profile estimation from wind profiler radar data is important for atmospheric studies and weather forecasting, yet it remains a challenging area due to low signal-to-noise ratio and other interferences. This project investigates and evaluates Doppler spectrum processing algorithms for the 205 MHz Stratosphere-Troposphere (ST) radar facility at the Cochin University of Science and Technology (CUSAT). Specifically, we analyse the Mutual Convolution Cost Function (MCCF) and Improved Moments Estimation (IME) algorithms, alongside hybrid approaches combining mutual convolution with IME pipelines.

To assess algorithmic performance without the limitations of conventional validation tools like radiosondes, a synthetic data test bench was developed by reversing the radar signal processing chain. Diverse raw data samples incorporating crude approximations of atmospheric wind phenomena, turbulence, and clutter were generated. The algorithms were evaluated using mean square error and Pearson's correlation coefficient. Simulation results demonstrate that the IME algorithm achieves superior accuracy and lower error rates compared to MCCF and other hybrid configurations.

List of contents

List of Figures ..... **ii**

List of Tables ..... **iii**

1 Introduction **1**

1.1 Wind profiler radar 1

1.2 The ST Radar Facility at CUSAT 2

1.3 Signal processing chain 3

1.4 Challenges in Wind Profiler Data Processing 3

1.5 Research motivation and problem definition 4

2 Estimation and Evaluation Methods **5**

2.1 Estimation Methods 5

2.1.1 Mutual Convolution Cost Function (MCCF) 5

2.1.2 Improved Moments Estimation (IME) 7

2.1.3 Hybrid Methods 10

2.2 Evaluation methods 11

2.2.1 Radiosonde 11

2.2.2 LIDAR 13

2.2.3 Synthetic Data and Validation 13

3 Results **16**

3.1 Evaluation with synthetic data 16

3.2 Evaluation with Radiosonde 18

3.3 Evaluation with Data Pattern's estimation algorithm 19

4 Future Scope **21**

References **22**

Appendix **23**

A.1 Source code 23

A.1.1 MCCF 23

A.1.2 IME 26

A.2 Presentation Slides 30

List of Figures

Figure 1.1 Doppler Beam Swing method 1
Figure 1.2 Signal processing chain 3
Figure 2.1 MCCF Algorithm Flow. 5
Figure 2.2 Power spectrum plots in MCCF 7
Figure 2.3 Preprocessing chain in IME Method 8
Figure 2.4 IME Adaptive window procedure 8
Figure 2.5 Grouping and profile tracing in IME 9
Figure 2.6 Power spectrum plots in IME 10
Figure 2.7 Flow of Hybrid-1 method 11
Figure 2.8 Flow of Hybrid-2 method 11
Figure 2.9 Radiosonde attached to weather balloon 11
Figure 2.10 Altitude reached by various radiosonde. 12
Figure 2.11 Trajectory of Radiosonde in the month of November 2023 12
Figure 2.12 Trajectory of Radiosonde in the month of February 2021 13
Figure 2.13 A Lidar mounted on the rooftop of a building 13
Figure 2.14 Various wind types and weather phenomena used in generating UVW array 15
Figure 2.15 Synthesised raw data and corresponding velocity estimation using MCCF 15
Figure 3.1 Comparison of mean MSE value of various algorithms in simulation . 16
Figure 3.2 Comparison of distribution MSE in various algorithms in simulation .17
Figure 3.3 Distribution of correlation between reference and computed value in simulated data 17
Figure 3.4 Correlation in each beam in simulated data 18
Figure 3.5 Projection of radiosonde observation along with wind velocity estimation using MCCF algorithm 18
Figure 3.6 Distribution of MSE between Data Pattern's moments and computed value in real-world data 19
Figure 3.7 Distribution of correlation between Data Pattern's moments and computed value in real-world data 19
Figure 3.8 Observation taken on 20 January 2026 11:00:41 IST 20
Figure 3.9 Observation taken on 20 January 2026 13:39:06 IST, estimation from MCCF 20

List of Tables

Table 1.1 Specification of ST Radar 2
Table 3.1 Mean Square error produced in each beam in simulation run 16
Table 3.2 Table of correlation values 17

Chapter 1
Introduction

The Earth's atmosphere is a chaotic, dynamic system with many phenomena happening. For a long time, researchers have been trying to model the atmosphere for weather forecasting and climate modelling. Among these, the wind-velocity gradient is important. In the past, the wind velocity was measured using radiosondes, a balloon-borne instrument released once now and then, at a fixed location. Although radiosondes remain the reference standard, they provide sparse temporal sampling, drift away from the launch site as they ascend, and cannot offer a continuous view of the evolving atmosphere. In this context, wind-profile radar provides a powerful complement.

1.1 Wind profiler radar

A wind profiler radar transmits electromagnetic pulses into the sky and picks up extremely weak echoes returned by minute fluctuations in the refractive index of air produced by turbulence and hydrometeors such as raindrops. Because the wind carries these scatterers, the Doppler shift of the returned signal can estimate the wind velocity along the radar beam. By pointing the beam in different directions, the 3D wind vector can be reconstructed at every range gate, giving a near-continuous record of the wind. The beams can be steered mechanically or electronically. This process is called Doppler Beam Swing (DBS).

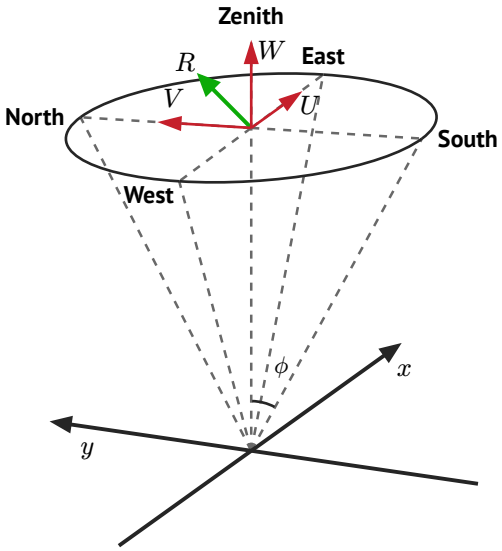


Figure 1.1: Doppler Beam Swing method

However, the quality of the record depends entirely on how reliably the wind velocity is estimated from the measured Doppler spectrum. In this report, we discuss a few algorithms used in velocity estimation, as well as a test bench upon which algorithms can be further evaluated and developed. The main focus is on processing radar data obtained from

Stratosphere–Troposphere radar (ST Radar) operating at the Cochin University of Science and Technology, Kochi.

1.2 The ST Radar Facility at CUSAT

The ST radar is located at the Advanced Centre for Atmospheric Radar Research (ACARR) on the university campus in Kochi (10°02′33.5″N 76°19′56.2″E). It was proposed to the Department of Science and Technology, Government of India, under its IRHPA scheme, with the specific aim of studying the characteristics and dynamics of the Indian summer monsoon at its gateway. Kerala is the point at which the south-west monsoon first makes landfall over the Indian subcontinent, and a continuously operating wind profiler at this location provides a unique vantage point on the onset, progress and intraseasonal variability of the monsoon circulation, as well as on other weather phenomena. The radar was designed, developed and installed indigenously, with the system fabricated by Data Patterns under the guidance of the university's scientists and engineers, and it became fully operational in December 2016. It is regarded as the first atmospheric wind profiler radar to operate successfully in the 205 MHz frequency band. The specifications of the ST radar are summarised in Table 1.1

Parameter	Specification
Location	10°02′33.5″N 76°19′56.2″E
Frequency	205 MHz
Bandwidth	5 MHz
Type of system	Active Phased Array
Antenna element	Three Element Yagi-Uda
Height coverage	315 m to 20 km
Range gates	1024 (Programmable)
Modes	DBS/SAM
Height resolution	$\approx$ 45 m to 720m
Beam width	$\approx 3°$
Off Zenith Angle	0° to 30° in steps of 1°
Beam positions in Azimuth	0° to 360° with 1° resolution
Pulse width	$0.3\mu s$ to $76.8\mu s$
Modulation	Binary Phase Shift Keying (BPSK)
Code	Complementary Code/Barker Code
Baud rate	$0.3\mu s$ to $4.8\mu s$ in steps of 0.3
PRF	100 Hz to 16 kHz Selectable
Peak Power	500 W (typical per element)

Parameter	Specification
Duty ratio	Up to 15 % (Max)
Peak power aperture product	$\approx 1.6 \times 10^8$ (typical) Wm^2
Radar system sensitivity	-165 dBm(post processing)
Dynamic range	70 dB(min)
Type of receiver	Direct band with sampling
Type of signal processor	FFT based frequency domain
Horizontal wind Velocity	Up to 70 ms^{-1}
Vertical wind velocity	Up to 30 ms^{-1}
Radial velocity resolution	Better than 0.1 ms^{-1}

Table 1.1: Specification of ST Radar from [1]

The radar has been validated using measurements from a radiosonde. The calibration is performed using the radio source Virgo A and by tracking the Moon

1.3 Signal processing chain

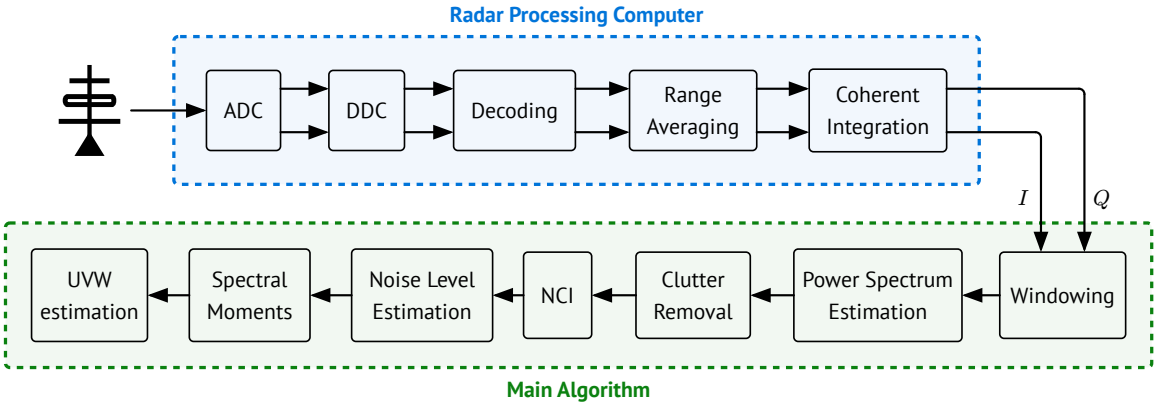


Figure 1.2: Signal processing chain

Complex In-phase and Quadrature (IQ) samples are collected for each range gate and beam direction; they are coherently integrated to improve the SNR. All these stages happen at the radar processing computer. Later, windowing is applied to reduce noise. Then the power spectrum is estimated via FFT; clutter algorithms are applied; Non-Coherent Integration (NCI) is applied; then the remaining noise level is estimated, which is useful in moment estimation. Finally, wind velocity in each beam direction and then UVW is computed.

1.4 Challenges in Wind Profiler Data Processing

Accurate wind estimation from wind profiler radar data is challenging because atmospheric echoes are generally weak and can be affected by noise, interference, and unwanted

reflections. Signal strength often decreases with altitude, making the atmospheric echo difficult to distinguish from background noise. Under low signal-to-noise ratio (SNR) conditions, identifying the correct Doppler peak becomes less reliable and may lead to inaccurate velocity estimates.

The received Doppler spectrum may also contain ground clutter, radio-frequency interference (RFI), and echoes from aircraft, birds, and insects. Ground clutter is generally concentrated around zero Doppler frequency and can mask atmospheric signals, while RFI and moving objects may introduce peaks that are unrelated to atmospheric variations. During rainfall, precipitation echoes can appear along with clear-air echoes, producing multiple peaks in the spectrum. Similarly, reflections from different atmospheric layers may result in overlapping spectral components.

These conditions make Doppler peak identification difficult, as the strongest peak does not always correspond to the actual atmospheric signal. In addition, wind profiler radars continuously generate large volumes of data across multiple beams and range gates. Therefore, the processing method must provide reliable estimates under varying atmospheric conditions while maintaining reasonable computational efficiency. Manual inspection of individual spectra is not practical for continuous operation, highlighting the need for robust and automated methods for Doppler spectrum processing and wind estimation.

1.5 Research motivation and problem definition

The challenges associated with Doppler spectrum processing highlight the need for reliable methods to identify atmospheric signals and estimate wind velocity accurately.

The main problem addressed in this work is the accurate identification of atmospheric Doppler peaks and estimation of their corresponding spectral moments from the radar observations. An incorrect peak selection can lead to errors in radial velocity estimation, which subsequently affect the retrieval of the three-dimensional wind components.

To address this problem, this project investigates the Improved Moments Estimation (IME) [2] and Mutual Convolution Cost Function (MCCF) [3] methods for Doppler spectrum processing. Their performance is evaluated using radar observations, with particular attention to peak identification accuracy. A hybrid approach combining the advantages of these methods is also tried to improve the reliability of wind estimation.

The overall aim is to develop and evaluate a suitable processing approach for the 205 MHz ST radar data that provides reliable Doppler spectral moments and wind profiles under different atmospheric conditions. During evaluation, we also introduce traditional methods like radiosonde observations and our own evaluation method based on synthetically generating radar data. A method refered as “Basic” is also included along with 4 other methods. It subtracts Hildreth-Seikh estimated noise level from spectrum and finds moments using standard form. We conclude our presentation by highlighting the need for synthetic data generation in algorithm evaluation.

Chapter 2
Estimation and Evaluation Methods

Various techniques for moment/peak estimations have been devised in the past decade. Some of them cover general cases while others are better suited for specific environments. In this chapter, we discuss a generic method and 2 established ones: MCCF and IME. We also discuss mixing IME and MCCF methods. In the 2nd part, we discuss some methods used to evaluate the accuracy of algorithms.

2.1 Estimation Methods

2.1.1 Mutual Convolution Cost Function (MCCF)

MCCF is a peak estimation algorithm proposed by Tang et al. [3] which uses Mutual convolution to improve SNR and a velocity-frequency-weighted cost function to estimate the most prominent peak in the spectrum. It also uses a space-time Doppler window (ST window) to limit the number of candidate peaks as well as to improve the continuity of the velocity profile. The space window is used to improve velocity continuity over height ranges, and the time window is used to improve velocity continuity in specific height over time.

The algorithm is illustrated in Figure 2.1. Here, the cylindrical entries are final data as well as intermediate data acquired. The intermediate data might be useful for other algorithms to work on in further processing steps.

Individual steps are explained in upcoming subsections. The method applied to a Doppler spectrum is shown in Figure 2.2

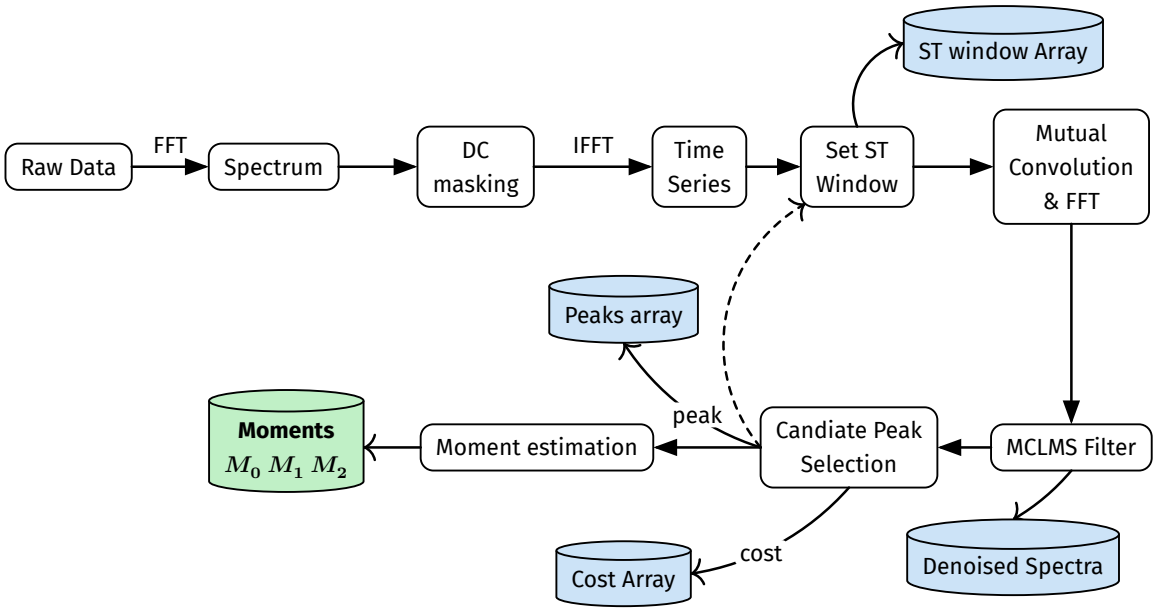


Figure 2.1: MCCF Algorithm Flow.

Preprocessing

Here the FFT of time series raw data (slow sampled) is computed. Then DC masking is applied, in which the values of 5 bins near DC bin is replaced with background noise with random phase. It is then converted back to time series data.

Setting the Space-Time Doppler Window

This step improves velocity profile continuity by limiting the number of candidate peaks. By setting a Doppler window in *space axis*, peaks outside this window are rejected. This window accounts for maximum wind shear between height bins. Setting a Doppler window in *time axis* applies the assumption that during a time interval, wind velocity changes at some maximum rate. The Space-Time window is the intersection of both windows.

Mutual Convolution

In this step, the noise is heavily suppressed, and SNR is improved. It works by slicing a spectrum in a height bin P into n slices of length $N/2$ (where N is length of original spectrum) then convolving adjacent slices ($X_i = V_i * V_{i+1}$) together to form $n-1$ results. From these results, an optimal solution segment X_r is chosen, which contains the highest peak among all results. Also, the segment adjacent to X_r is chosen as the suboptimal solution X'_r. Both of these are used in the MCLMS filter.

MCLMS Adaptive Filtering

MCLMS (Modified Complex Least Mean Square) filter is an adaptive filter. It learns its weights during slicing of spectra, and after modifying the weight vector, the final filtered spectra is obtained through the optimal and suboptimal spectra. MCLMS can suppress clutter even more by training a model that tries to match the optimal solution with the suboptimal solution. In the end, the spectrum will have some of the clutter suppressed.

Candidate Peak Selection, Cost Function & Moment Estimation

Peaks in ST Doppler window which has amplitude $\geq 40\%$ of maximum peak are chosen as candidate peaks. The algorithm assigns a "cost" to each of these peaks based on their velocity difference between each frequency bin $V_{r+1,j}$ given by:

$$V_{r+1,j} = 1 - \left| \left(M_{r+1,j} - M_r\right) \frac{d_v}{\alpha V_d} \right| \tag{2.1}$$

and magnitude of peak itself $M_{r+1,j}$. Here d_v represents the velocity resolution, V_d represents the maximum wind shear, α is the window coefficient. The cost of each peak is given by:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \tag{2.2}$$

The peak with the highest cost value is taken as the spectral peak. From this, moments can be estimated based on the Hildreth-Seikh method.

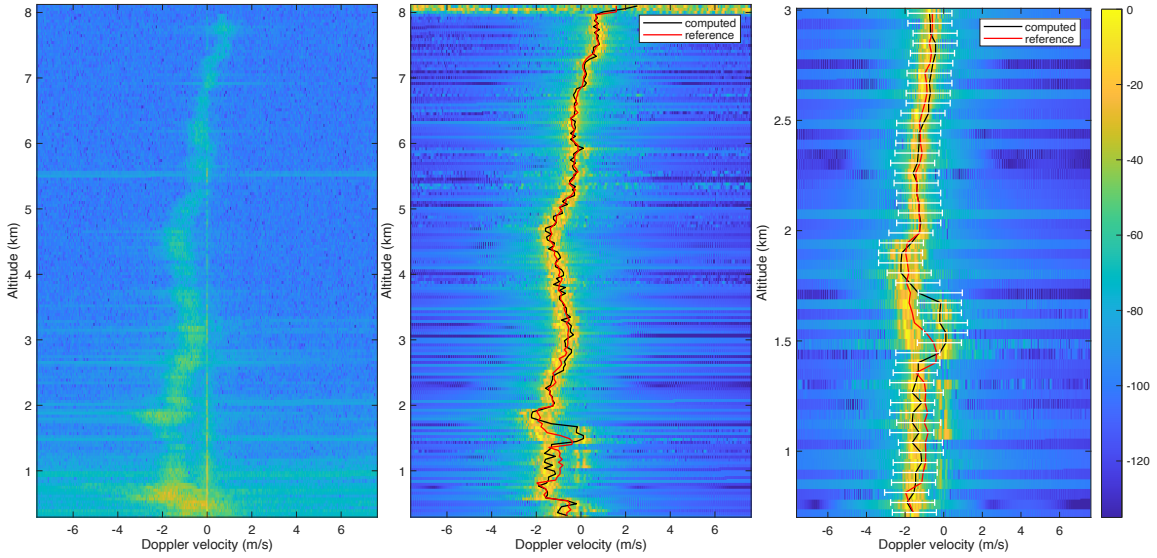


*Figure 2.2: Power spectrum (a) Original, (b) after applying DC removal, mutual convolution, and MCLMS, (c) Computed moments, reference moment and **Space window [white]** overlaid*

The peaks estimated from one height bin are used to set the ST window in the next height bin; hence the process is iterative and cannot be done in a single matrix operation step as in computing a spectrum.

2.1.2 Improved Moments Estimation (IME)

The Improved Moments Estimation (IME) algorithm proposed by Li **et al.**[2] is designed to extract reliable Doppler velocity profiles from ground-based K-band wind radar spectra, particularly when the atmospheric echoes have low SNR and are contaminated by clutter or multiple spectral peaks. The method combines the adaptive Doppler-window concept of the Adaptive Moments Estimation (AME) method with the cost-function-based profile tracing concept of the Multiparameter Cost Function (MPCF) method. Its main objective is to reduce the computational search space while maintaining reliable wind-profile detection under low-SNR conditions.

The IME procedure consists of three main stages:

1. Preprocessing of the Doppler power spectra
2. Adaptive Doppler-window selection and identification of prospective spectral peaks
3. Grouping, profile tracing, and cost allocation.

Data Preprocessing

Before profile extraction, the Doppler power spectra are processed to suppress unwanted spectral components. First, the zero-frequency region is affected by direct-current (DC) components and ground clutter. These components are suppressed by interpolation, in which the average values of three points around zero frequency are used to replace the

contaminated values. Next, a five-point moving-average filter is applied to suppress transient interference while preserving the atmospheric echo characteristics. Finally, the noise power density S_n of each range bin is estimated using the Hildebrand–Sekhon method. The estimated noise level is subtracted from the power spectrum, and negative values are set to zero. The resulting spectrum is therefore more suitable for identifying atmospheric spectral peaks. The preprocessing procedure described in the paper is summarised in Figure 2.3

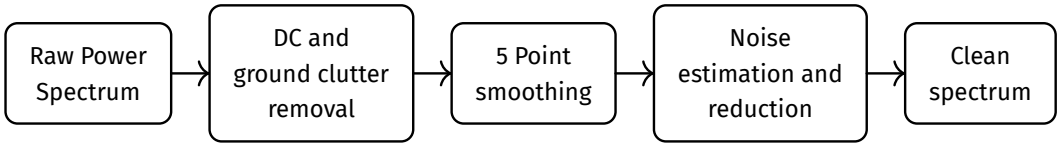


Figure 2.3: Preprocessing chain in IME Method

These preprocessing operations help identify weak atmospheric echoes by reducing the effects of random noise and radio-frequency interference.

Adaptive Doppler Window and Prospective Peak Selection

The IME algorithm starts at the lowest range bin and searches for candidate spectral peaks only within an adaptive Doppler window. For a selected spectral component, the algorithm determines the peak position and the left and right zero-crossing points, from which the radial velocity, spectral width, and SNR are calculated. Up to three peaks are retained in each range bin as a prospective atmospheric component when each of which having SNR is greater than 0 dB. No peaks are selected in a range bin if there are no spectral component with power greater than the estimated noise level in the current height.

For the first range bin, no previous information is available. Therefore, the most prominent spectral component is initially selected, and its radial velocity is used as the mean Doppler velocity. The Doppler window for the next range bin is then centred around the candidate whose radial velocity is closest to the previous mean Doppler velocity. This process is repeated sequentially with increasing range. This procedure is illustrated in Figure 2.4

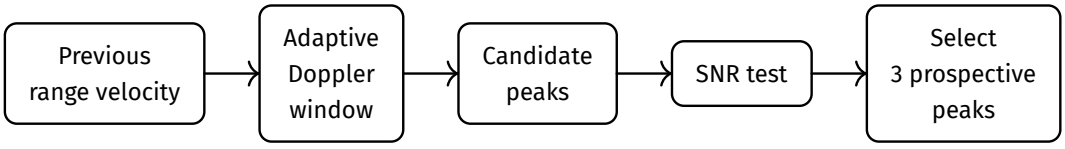


Figure 2.4: IME Adaptive window procedure

The wind Doppler signal is generally not expected to change by more than 20% of the total coherent-filter bandwidth, and therefore the Doppler-window width is restricted to less than 20% of the complete Doppler-axis width.

Grouping and Cost-Function-Based Profile Tracing

In this step, the algorithm constructs possible traces between adjacent range bins. A trace represents a possible Doppler velocity profile through height. Constructing all pos-

sible traces over the complete range would result in a large computational load; therefore, the spectra are divided into small groups containing at least three consecutive range bins. Candidate peaks within each group are then connected from one range bin to the next. For the first group, the initial value is the Doppler velocity of the most prominent peak, while subsequent groups use the velocity at the highest range bin of the preceding group as their initial value. Figure 2.5 shows all links in a group and M_j^r indicates the j th peak of the r th range bin.

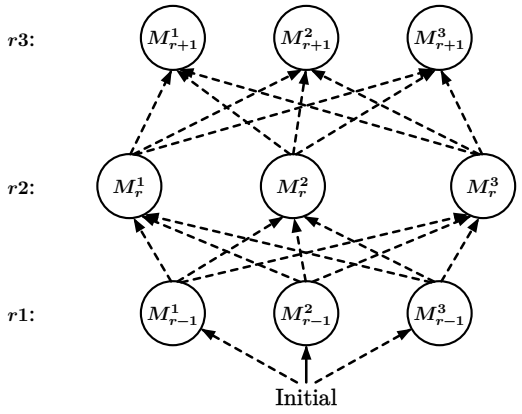


Figure 2.5: Grouping and profile tracing in IME. Each circle represents a candidate spectral peak

The cost assigned to a connection between two candidate peaks is based on two characteristics: the continuity of wind shear and the relative SNR of the candidate. The maximum allowable radial-velocity shear is determined by;

$$|\Delta v_{\mathrm{m}}| \leq 0.05 \sin(\theta) \Delta R \tag{2.3}$$

where Δv_{m} is the maximum radial-velocity shear, θ is the radar beam tilt angle, and ΔR is the range resolution.

To represent the continuity of the Doppler profile, a trapezoidal membership function is used:

$$T(x) = \begin{cases} 1 & |x| \leq \Delta v_{\mathrm{m}} \\ (2\Delta v_{\mathrm{m}} - |x|)/\Delta v_{\mathrm{m}} & \Delta v_{\mathrm{m}} < |x| < 2\Delta v_{\mathrm{m}} \\ 0 & |x| \geq 2\Delta v_{\mathrm{m}} \end{cases} \tag{2.4}$$

Here, x represents the wind-shear difference between two connected candidate peaks. A high membership value indicates that the change in velocity between adjacent range bins is consistent with the expected continuity of the atmospheric wind profile.

The relative SNR of a candidate peak is normalised with respect to the largest candidate SNR in the same range bin:

$$P_{r,j} = \frac{\alpha_{r,j}}{\max(\alpha_r)} \tag{2.5}$$

where $\alpha_{r,j}$ denotes the SNR of the j-th candidate peak in the r-th range bin. The complete link cost is then formed by combining the continuity term and the normalised SNR term:

$$C = w_1 T + w_2 P, \qquad w_1 + w_2 = 1. \tag{2.6}$$

The weights determine the relative importance of wind-profile continuity and spectral strength. Based on experiments as described in the research paper [2], the selected values are,

$$w_1 = 0.4, \quad w_2 = 0.6. \tag{2.7}$$

Thus, the IME algorithm gives greater weight to the normalised SNR while still enforcing continuity between adjacent range bins. The total cost of each possible trace is calculated, and the trace with the highest total cost is selected as the final Doppler velocity profile for the corresponding group. The velocity at the highest range bin in the selected group then becomes the initial value for the next group.

Figure 2.6 illustrates how IME narrows the peak-search region and identifies a more consistent atmospheric Doppler profile.

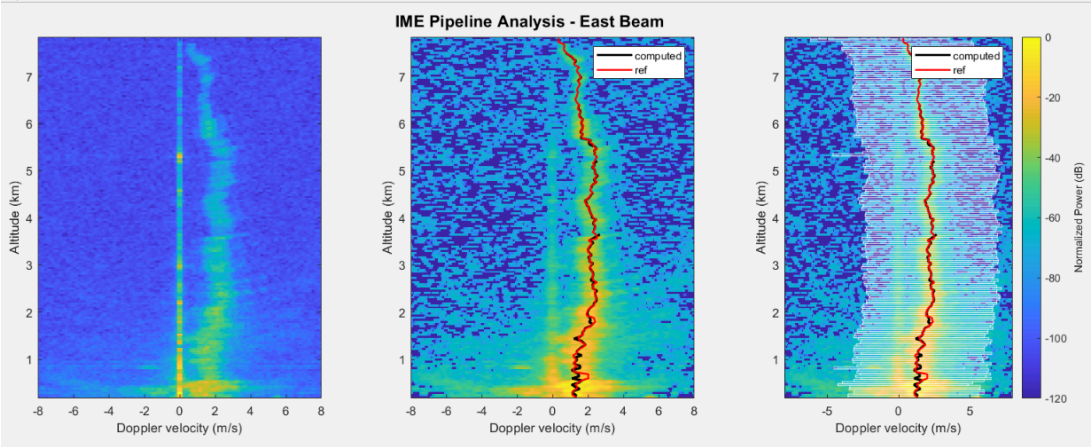


Figure 2.6: Doppler beam spectra: a) Original, b) after IME processing, c) Adaptive Doppler-window selection.

2.1.3 Hybrid Methods

The 2 algorithms discussed have several individual working components – computing spectrum, denoising, estimating peaks/moments, etc. The mutual convolution process in MCCF appears to suppress noise very effectively. In this light, we tried a method that combines both of these methods.

To make such mixing easier, our main focus was to first create a software architecture in MATLAB that encapsulates each of these components and provides a common interface. Then the individual components can be cascaded easily. Figure 2.7 Shows the method in which Mutual Convolution (MC) from MCCF is first used to determine the spectrum, and then later processing is done by IME itself.

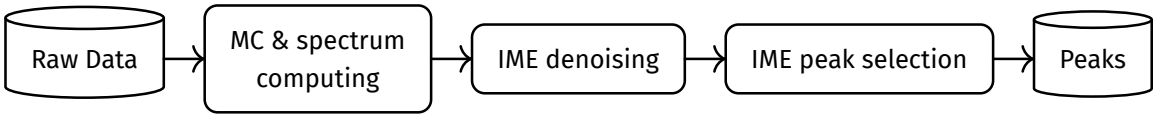


Figure 2.7: Flow of Hybrid-1 method

Figure 2.8 shows another method in which we eliminated the denoising part of IME (since MC already performs some amount of denoising, we suspect that further denoising eliminates candidate peaks in low SNR regions)

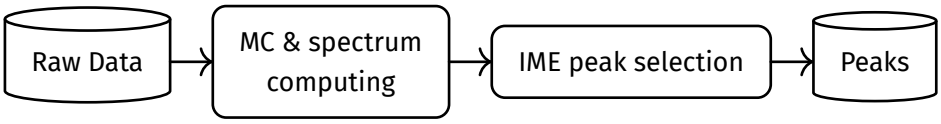


Figure 2.8: Flow of Hybrid-2 method

In practice, the parts of MCCF work iteratively in each height bin, taking feedback from the previous bin, setting the ST window appropriately, then selecting the best solution from MC that contains a peak falling in the ST window. For this mixing purpose, we took the MC part alone and let it run without the feedback.

2.2 Evaluation methods

To evaluate the algorithms, we need a reliable ground truth to compare against the output of the algorithm. Below are a few methods generally used in the literature, along with our proposed method: synthetic data.

2.2.1 Radiosonde

A radiosonde is a battery-powered telemetry instrument carried into the atmosphere, usually by a weather balloon, that measures various atmospheric parameters and transmits them by radio to a ground receiver. Modern radiosondes measure or calculate the following variables: altitude, pressure, temperature, relative humidity, wind (both speed and direction), and some are also capable of cosmic ray readings at high altitude.



Figure 2.9: Radiosonde attached to weather balloon

The radiosonde reaches an altitude of 20 km to 30 km; thereafter, due to the low density of the atmosphere, the balloon explodes, and it falls down. The altitude reached in various radiosonde experiments is plotted in Figure 2.10

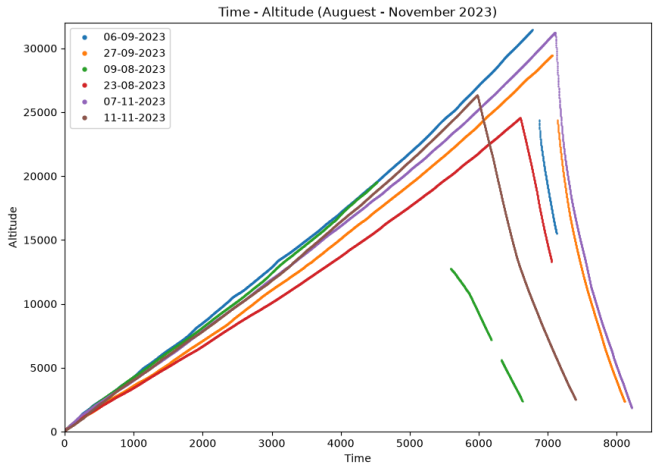


Figure 2.10: Altitude reached by various radiosonde.

In order for the data from the radiosonde to be useful for evaluating the algorithm, the balloon must hover over the same geographic area throughout the desired height range. For perspective, the ST radar has a beamwidth of 3°, Considering 30km target height, ideally, the ballon should be within $\tan(3°/2) \times 30 \text{ km} \approx 1 \text{ km}$ radius within the launch site. But this can be much higher, considering that the atmosphere can be slowly varying in certain seasons. However, the radiosondes often drift too far from their initial destinations. This is illustrated in Figure 2.12 and Figure 2.11. Data collected in the summer season are used for testing the algorithm. However, in those seasons, wind velocities are not turbulent enough to rigorously test the algorithm.

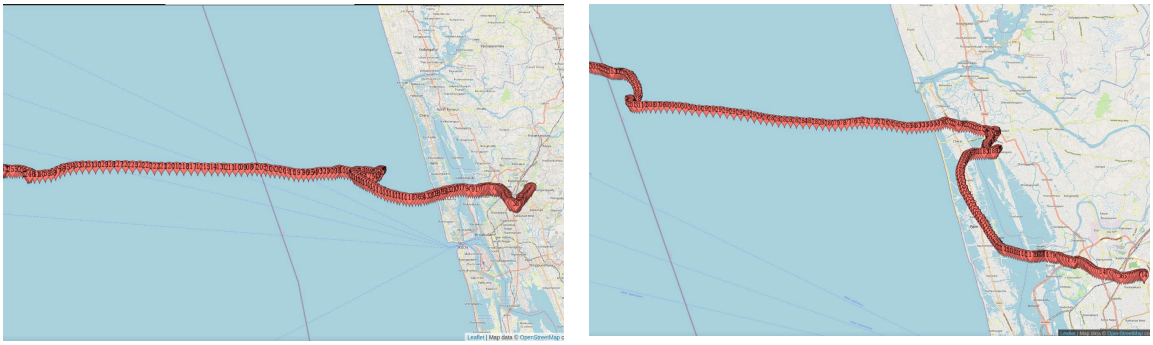


Figure 2.11: Trajectory of Radiosonde in the month of November 2023

Moreover, radiosondes are rarely launched at intervals of less than 6 hours and so can only be used to provide information about the large-scale weather patterns.

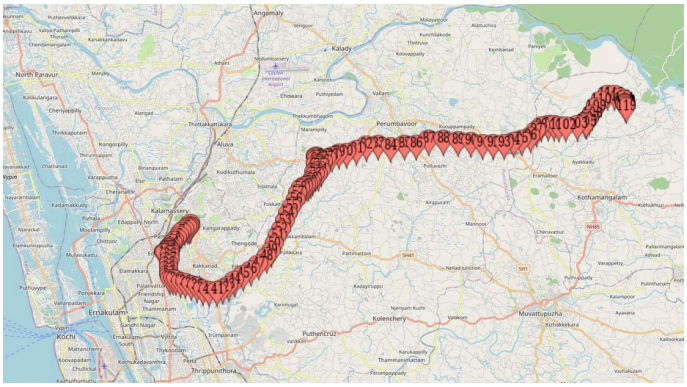


Figure 2.12: Trajectory of Radiosonde in the month of February 2021

2.2.2 LIDAR

Light Detection and Ranging (LIDAR) is a method for determining ranges by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the optical receiver. Lidar may operate in a fixed direction (e.g., vertical), or it may scan directions, in a special combination of 3D scanning and laser scanning. In [3] and [2], the researchers used a ground-based lidar for measuring the wind velocity of the lower atmosphere (up to 1 km). In [3], it is mentioned that the lidar can achieve 99% accuracy in clear atmospheric conditions. Hence, lidar serves as an excellent ground reference for evaluating the algorithm's performance.



Figure 2.13: A Lidar mounted on the rooftop of a building

2.2.3 Synthetic Data and Validation

Synthetic radar data are artificially generated radar observation data made from a predefined array of wind velocities. This is done by starting with a UVW array and reversing the process of a typical radar signal processing pipeline, starting from UVW estimation to writing a raw file onto disk. The primary advantage of this method is that we have a ground truth to compare the algorithm's output, as well as control to create data that can test the algorithms in various ways, which otherwise would require waiting to get suitable outputs from radar and manually going through them to pick a good one.

Background

While the algorithms are being developed, it is important to measure their performance with some metric. To validate the algorithm, a ground truth is required. Traditionally, the default output of the radar signal processing chain implemented by Data Patterns was used. This approach has some drawbacks: first, the original algorithm may fail under certain breaking conditions (poor SNR, wind discontinuity, heavy interference, etc.), so it yields no useful data points in those height bins. Second, to test the algorithm under certain weather conditions, one must collect real-world observations, which may or may not be catalogued by weather patterns. Hence, for large-scale analysis of algorithms, reliance on physical data might not be viable. From this, we propose an idea of synthesising radar data on our own according to the specific needs.

Method of synthesis

The synthesiser is a reverse process of the radar processing chain shown in Figure 1.2. First, a UVW array is synthesised based on the needs of atmospheric conditions. It is generated by combining wind velocities generated by various types of winds and atmospheric phenomena. Currently, we took 8 different processes as shown in Figure 2.14. Then velocities are projected onto each beam direction; velocities are converted to a Doppler spectrum with spectrum broadening using [4] and Zrnić's method [5]. Additional clutters like ground and bright bands are added to the final spectrum.

The resulting spectrum is converted to the time domain. To create a radar raw file, a dummy header file that has a pattern similar to that found in actual radar is created, and this time series data is attached to it. The synthesised file is opened again with the same radar processing chain to test its validity. One sample of synthesised raw data is shown in Figure 2.15. The plot also compares the actual wind velocity with the output of the MCCF function. One important detail to note here is that not all components represent the actual physical processes. For eg, the synoptic background profile uses a basic model of wind velocity gradient due to surface friction. We used AI to figure out an initial list of wind components. However, our goal was to get a reasonably continuous, smooth pseudorandom wind profile and not to simulate the atmosphere as realistically as possible.

We created a testbench that generates 500 raw data samples, each containing beams in 5 directions. For each data sample, 5 different algorithms were applied; their MSE and correlation coefficient were computed, stored in a file and then processed later. The results are shown in the results chapter.

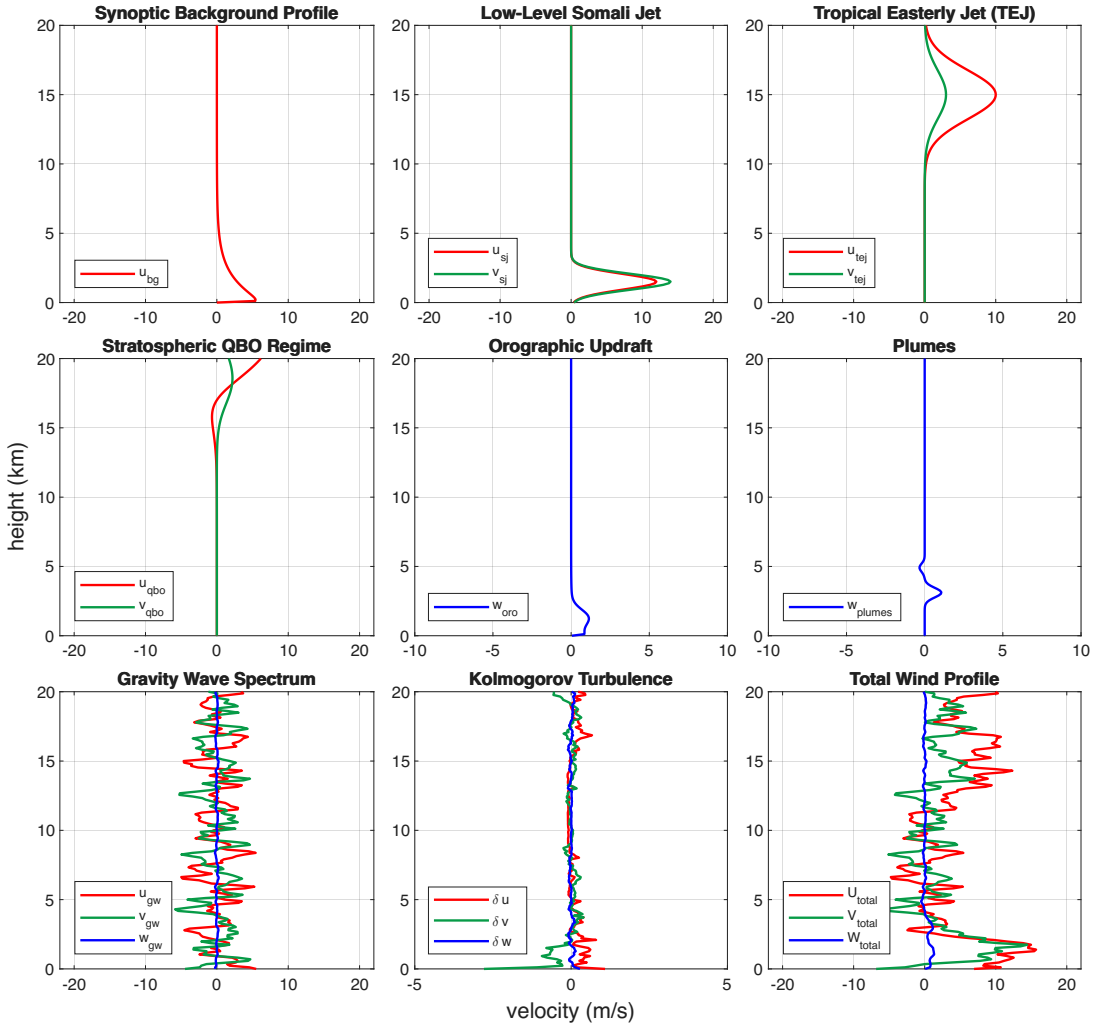


Figure 2.14: Various wind types and weather phenomena used in generating UVW array

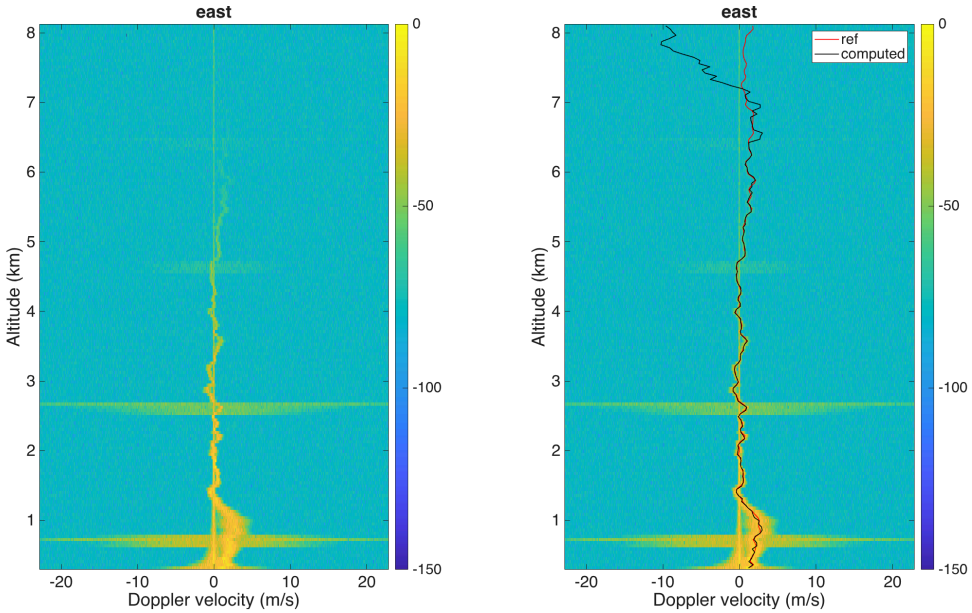


Figure 2.15: Synthesised raw data and corresponding velocity estimation using MCCF

Chapter 3
Results

To test the accuracy of methods, many metrics can be used, such as latency, computational efficiency, accuracy, etc. For ST radar, which completes a full observation across 5 beams every 80 seconds for wind profiling, the first two seem irrelevant. Hence, to measure accuracy, we have used Mean Square Error (MSE) and Pearson's linear correlation [6]. Here we present evaluation with synthetic data and real data. Since IME produces discontinuities in certain cases, a fair evaluation of algorithms are difficult. If we set zero value for all height bins where IME is not able to find velocity, it systematically gets much lower MSE (as in higher altitudes, wind tend to be much slower). Hence throughout the analysis, MSE and correlation of each algorithm is computed only in height ranges in which IME has a defined output.

3.1 Evaluation with synthetic data

In this method of evaluation, 500 raw data points were synthesised. Each of which was produced from the UVW array synthesised using randomised parameters. Additionally, ground clutter and bright bands at random altitudes were added to the spectrum. The mean MSE in each of the beams are shown in Figure 3.1. The distribution of MSE is summarised in Figure 3.2. Table 3.1 shows exact values.

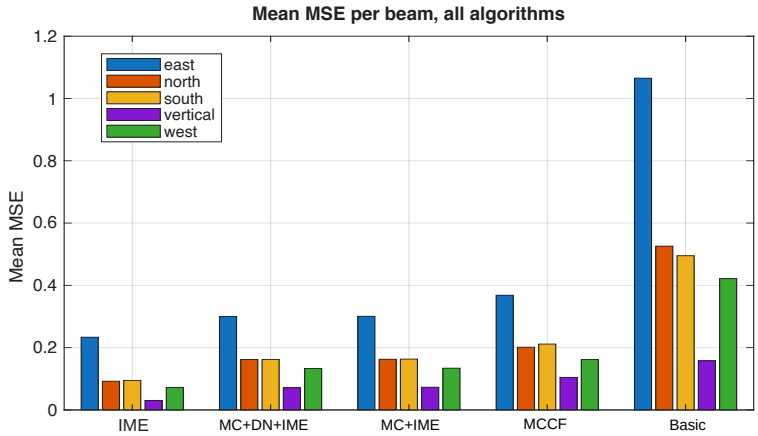


Figure 3.1: Comparison of mean MSE value of various algorithms in simulation

	IME	MC+DN+IME	MC+IME	MCCF	Basic
East	0.233	0.299	0.300	0.368	1.065
North	0.092	0.161	0.162	0.201	0.525
South	0.094	0.161	0.163	0.211	0.495
Vertical	0.030	0.071	0.073	0.104	0.157
West	0.072	0.133	0.134	0.161	0.421

Table 3.1: Mean Square error produced in each beam in simulation run

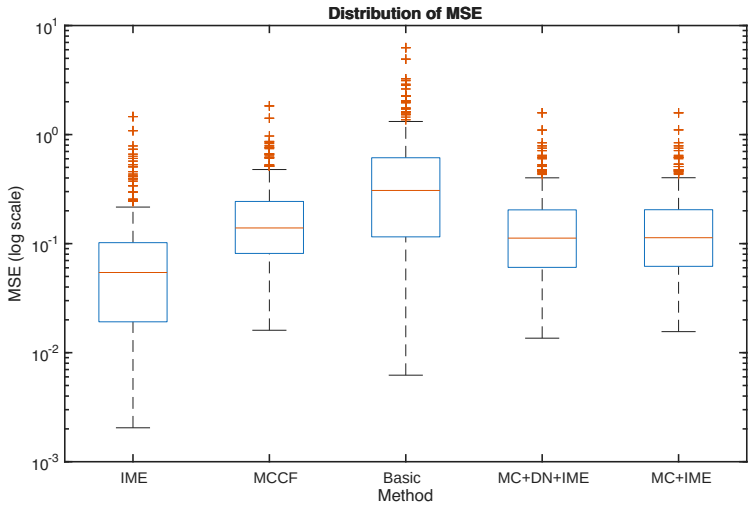


Figure 3.2: Comparison of distribution MSE in various algorithms in simulation

From Figure 3.1, it is clear that IME has better accuracy amongst all methods. The least error in the vertical direction is expected because in that direction, wind velocity is close to zero most of the time. However, the high error in the east beam is unexpected, and we couldn't trace any specific cause for it. From Figure 3.2, the distribution of MSE is much more spread out in the basic method. The hybrid method shows no difference and is suboptimal compared to IME, indicating that the combination of IME and mutual convolution is inefficient compared to IME alone.

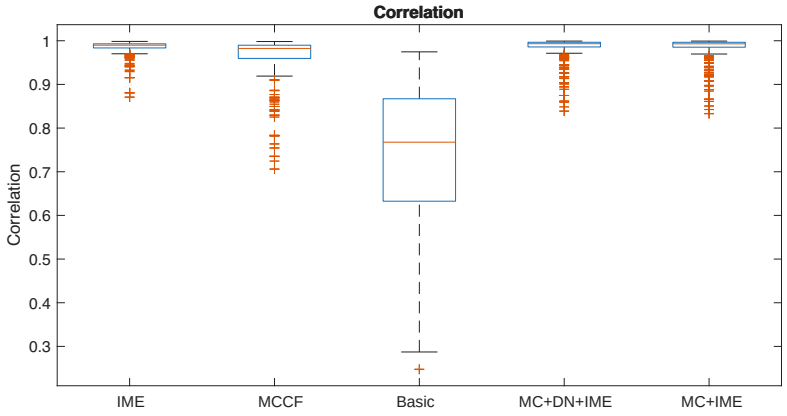


Figure 3.3: Distribution of correlation between reference and computed value in simulated data

	IME	MC+DN+IME	MC+IME	MCCM	Basic
East	0.987	0.992	0.992	0.984	0.701
North	0.988	0.990	0.990	0.978	0.732
South	0.985	0.987	0.987	0.968	0.750
Vertical	0.977	0.958	0.955	0.913	0.752
West	0.987	0.990	0.989	0.973	0.761

Table 3.2: Table of correlation values

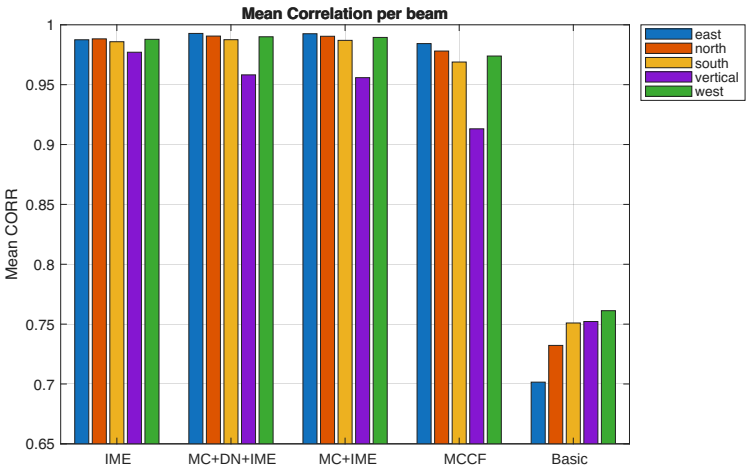


Figure 3.4: Correlation in each beam in simulated data

The Pearson's linear correlation between computed wind velocity and the reference is plotted in Figure 3.3 and Figure 3.4, the exact values are given in Table 3.2. The correlation turns out to be a poor measure compared to MSE. This is partly due to the scale invariance of the correlation function. However, the correlation is used in the literature. Hence, we decided to include it in this report. Paradoxically, here, the east beam has the highest correlation among all beams (except in basic algorithm). The difference is minute, which can result from slight numerical error in the computation flow.

3.2 Evaluation with Radiosonde

Matching data with radiosonde was a difficult task as it requires stitching different raw data. This is because a radiosonde covers 9km in about 30 minutes; hence, it smears the wind velocity. Moreover, the radiosonde is only able to measure horizontal wind velocity.

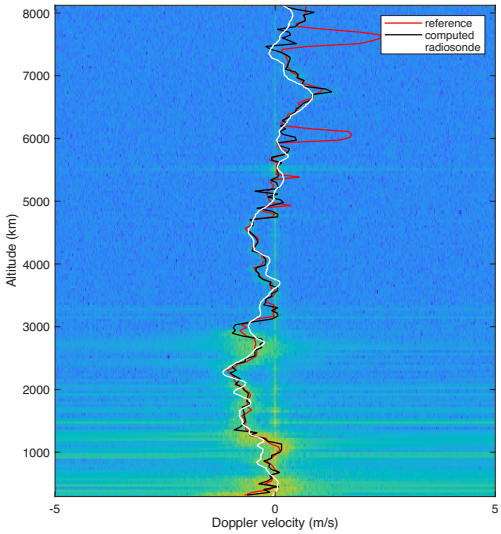


Figure 3.5: Projection of radiosonde observation along with wind velocity estimation using MCCF algorithm

However, the radar observation in cardinal directions contains both horizontal and vertical wind components; hence, matching them is very difficult. Figure 3.5 shows the projection of radiosonde observations on the north beam. The radiosonde was launched on 20th June 2026 at 14:20:25 IST. The radar data shown here was taken on the same day at 14:22:11 IST. Being taken on a season which has stable atmosphere, the radiosonde measurement matches the one with radar observation to some extend. The time-averaged velocity obtained from radar can be compared.

3.3 Evaluation with Data Pattern's estimation algorithm

226 radar observations taken on 20 January 2026 were analysed in batch similar to how it was done on synthetic data. The radar data was fed to 5 different algorithms, and their MSE and correlation were computed with respect to moments estimated by Data Pattern's algorithm. The results are shown in Figure 3.6 and Figure 3.7

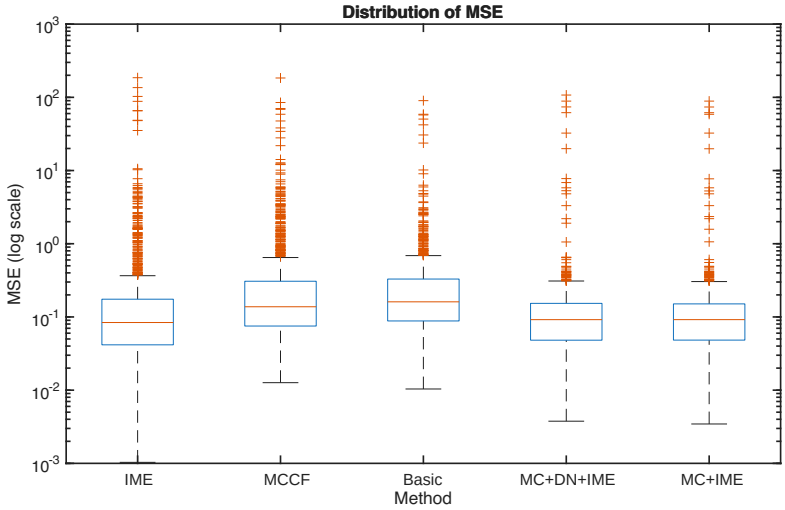


Figure 3.6: Distribution of MSE between Data Pattern's moments and computed value in real-world data

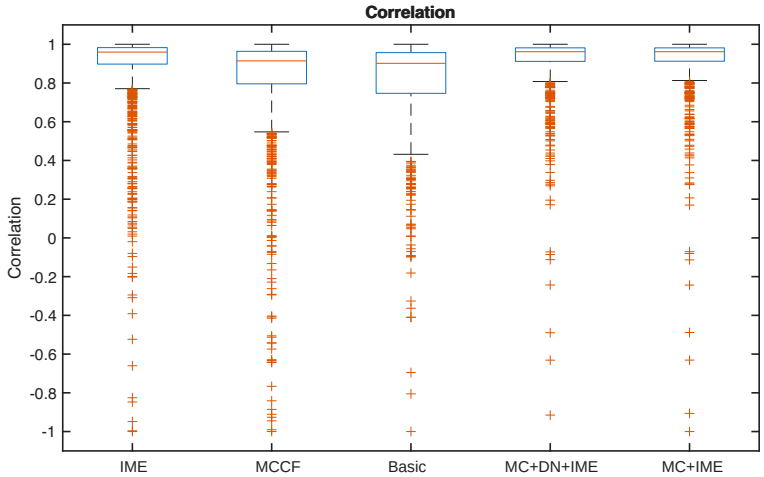


Figure 3.7: Distribution of correlation between Data Pattern's moments and computed value in real-world data

In this case, MC+IME method was marginally better than other methods. However, the Data Pattern's algorithm is not robust in all scenarios. One such case is shown in Figure 3.8, showing heavy deviation from normal. We also found certain cases where IME didn't produce any estimate, returning an empty array and forcing the `corr` function to return -1.

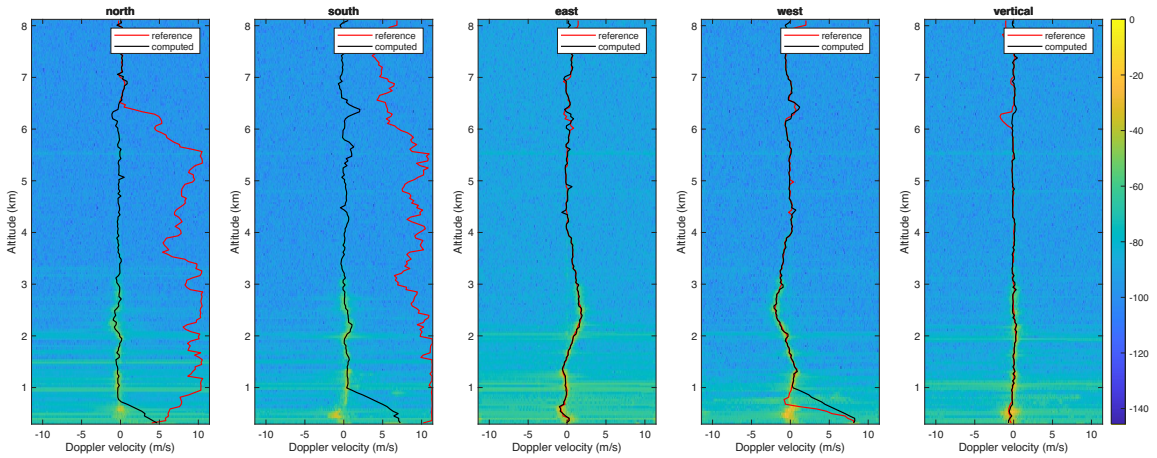


Figure 3.8: Observation taken on 20 January 2026 11:00:41 IST. The red plot shows the Data Pattern algorithm's estimate, and the black plot is MCCF's estimate

In this observation, the estimate of moments in Meridional directions by Data Pattern's algorithm was extremely off from normal estimates. However, neither estimation from our algorithms was perfect, as shown in Figure 3.9. Here, both algorithms fell short because the initial estimate of wind velocity in the first height bin was wrong. Perhaps an algorithm that takes different approach needs to be developed.

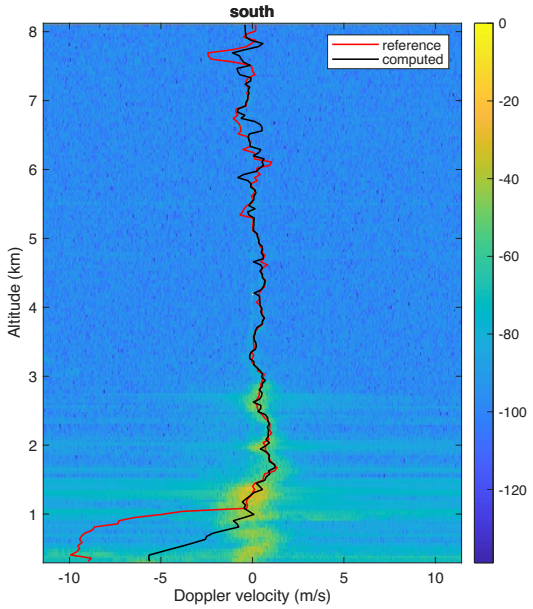


Figure 3.9: Observation taken on 20 January 2026 13:39:06 IST, estimation from MCCF

These issues highlight the requirement for a synthetic data-driven testing framework.

Chapter 4
Future Scope

The existing work can be extended further in many ways. Below is a list of a few of them;

1. **Wavelet-Based Denoising**: Wavelet denoising is a signal processing technique that removes unwanted background noise while preserving sharp edges, spikes, and fine details. The method could improve weak Doppler peak detection under low-SNR conditions.
2. **ML-Based Peak Detection**: Machine learning-based methods can be applied to improve identification of suitable peaks in multi-peak scenarios [7]. The approach can be further extended to separate rain echo from the spectrum.
3. Making synthesised radar data more realistic by adding more features. The current implementation still looks artificial. One critical issue with the current approach is that the signal is always detectable in lower height bins, due to how background noise is generated.
4. Distinguish rain echoes from clear-air atmospheric echoes in Doppler spectra to improve the estimation of wind velocity.
5. Synthetic data creation can be used to fill gaps between observations in the lower atmosphere while the experiment is being conducted at higher altitudes. However, this needs extensive modelling of the local atmosphere.
6. Sufficiently realistic synthetic data can be used to train reinforcement learning-based algorithms in estimating wind profiles.

References

[1] Titu K. Samson, "Design, Development, Calibration and Validation of 205 MHz Wind Profiler Radar," 2019. [Online]. Available: http://hdl.handle.net/10603/313988

[2] Li et al., "Improved Moments Estimation for Ground-Based K-Band Doppler Radar Using Cost Function Method," *IEEE Access*, vol. 10, Mar. 2022.

[3] Zaixing Tan et al., "Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method," *IEEE Sensors Journal*, vol. 24, no. 15, Aug. 2024.

[4] Fred E. Nathanson, *Radar Design Principles.* [Online]. Available: https://selasl.wordpress.com/wp-content/uploads/2011/08/mcgraw_hill_-_radar_design_principles.pdf

[5] Dušan S. Zrnić, "Simulation of Weatherlike Doppler Spectra and Signals," *Journal of Applied Meteorology and Climatology*, vol. 14, Jun. 1975.

[6] "corr - Linear or rank correlation - MATLAB." [Online]. Available: https://in.mathworks.com/help/stats/corr.html

[7] Zheng Zhang et al., "Radar Signal Recognition Based on CSRDNN Network," *IEEE Access*, vol. 12, Jun. 2024.

Appendix

A.1 Source code

Entire MATLAB implementations of the algorithms are available in
https://github.com/roopeshor/Radar-Signal-Processing

Below is the implementation of MCCF and IME

A.1.1 MCCF

```
function new_beam = compute_all_params(beam, cfg)
arguments (Input)
  beam RadarData
  cfg.prev_frame_peaks (1,:) double = []
    cfg.apply_mclms (1,1) logical = true
    cfg.win_coeff (1,1) double = 2
    cfg.wind_shear (1,1) double = 1
    cfg.velocity_res (1,1) double = 0.06
    cfg.cost_weights (1,2) double = [0.5, 0.5]
end
arguments (Output)
  new_beam RadarData
end
beamData = beam.BeamData;
[height_bins, nfft, ~] = size(beamData);
alpha =cfg.win_coeff;
max_shear =cfg.wind_shear;
dv =cfg.velocity_res;
if (isfield(beam.a_params, "alpha"))
  alpha = beam.a_params.alpha;
end
if (isfield(beam.a_params, "max_shear"))
  max_shear = beam.a_params.max_shear;
end
if (isfield(beam.a_params, "velocity_res"))
  dv = beam.a_params.velocity_res;
end
df = round(max_shear / dv); w1 = cfg.cost_weights(1);
w2 = cfg.cost_weights(2);
mclms_len = 10;
cost_scores = zeros(height_bins, nfft);
mclms_denoised_spectra = zeros(height_bins, nfft);
peaks_idx = zeros(height_bins, 1);
F0 = [];
spectra = fft(beam.BeamData, [], 2);
spectra = mean(spectra, 3);
noise_val = median(abs(spectra), 2);
[height_bins, nfft] = size(spectra);
mask_bins = [1, 2, 3, nfft, nfft-1];
```

```
rand_phase = exp(1j * 2 * pi * rand(height_bins, length(mask_bins)));
spectra(:, mask_bins) = noise_val .* rand_phase;
CS = ifft(spectra / max(spectra(:)), [], 2);
ST_W_array = zeros(height_bins, 2);
for r = 1:height_bins
  signal_clean = CS(r, :);
        if r == 1
        ST_W = 1:nfft;
  else
        W1 = max(1, round(F0 - alpha*df)) : min(nfft, round(F0 + alpha*df));
    if ~isempty(cfg.prev_frame_peaks) && length(cfg.prev_frame_peaks) >= r
      T0 = cfg.prev_frame_peaks(r);
            W2 = max(1, round(T0 - alpha*df)) : min(nfft, round(T0 +
alpha*df));
      inter_W = intersect(W1, W2);
      if length(inter_W) >= alpha * df
        ST_W = inter_W;
      else
        ST_W = W1;
      end
    else
      ST_W = W1;
    end
  end
  ST_W_array(r, :) = [min(ST_W), max(ST_W)];

  seg_len = floor(nfft / 2);
  num_delays = 8;
  step = floor((nfft - seg_len) / (num_delays - 1));
  v_r1 = signal_clean(1:seg_len);
  V_conv = zeros(num_delays - 1, seg_len);
  max_powers = zeros(1, num_delays - 1);
  for d = 2:num_delays
    start_idx = 1 + (d-1)*step;
    v_ri = signal_clean(start_idx : start_idx + seg_len - 1);
    v_ri_hr = conj(v_ri(end:-1:1));
    v_ci = conv(v_r1, v_ri_hr);
        v_ci_center = v_ci(floor(seg_len/2) : floor(3*seg_len/2) - 1);
    V_conv(d-1, :) = v_ci_center;
    spec_d = abs(fftshift(fft(v_ci_center, nfft))).^2;
        max_powers(d-1) = max(spec_d(ST_W));
  end
  [~, best_d_idx] = max(max_powers);
    X_opt = V_conv(best_d_idx, :);
    if best_d_idx > 1
    X_sopt = V_conv(best_d_idx - 1, :);
  else
    X_sopt = V_conv(best_d_idx + 1, :);
  end
        Y_r = X_opt;
  if (cfg.apply_mclms)
    Y_r = mccf.mclms_filter(X_opt, X_sopt, mclms_len);
  end
  spec_mclms = abs(fftshift(fft(Y_r, nfft))).^2;
```

```
  spec_mclms = spec_mclms / max(spec_mclms);
  mclms_denoised_spectra(r, :) = spec_mclms;
        if r == 1
        avg_conv = mean(V_conv, 1);
    final_spectr = abs(fftshift(fft(avg_conv, nfft))).^2;
    [~, selected_peak_idx] = max(final_spectr);
  else
    ST_W_spec = spec_mclms(ST_W);
    M_rmax = max(ST_W_spec);
    [pks, locs_in_stw] = findpeaks(ST_W_spec);
        valid_mask = pks >= 0.4 * M_rmax;
    cand_pks = pks(valid_mask);
    cand_locs = ST_W(locs_in_stw(valid_mask));
    if isempty(cand_locs)
      [~, max_rel_idx] = max(ST_W_spec);
      cand_locs = ST_W(max_rel_idx);
      cand_pks = M_rmax;
    end
        [cand_pks, sort_idx] = sort(cand_pks, 'descend');
    if length(cand_locs) > 3
      cand_locs = cand_locs(sort_idx(1:3));
      cand_pks = cand_pks(sort_idx(1:3));
    end
    best_cost = -inf;
    selected_peak_idx = cand_locs(1);
    for c = 1:length(cand_locs)
      cand_idx = cand_locs(c);
      cand_power = cand_pks(c);
            V_cost = 1 - abs((cand_idx - F0) * dv) / (alpha * max_shear);
            cost_val = w1 * V_cost + w2 * cand_power;
      cost_scores(r, cand_idx) = cost_val;
      if cost_val > best_cost
        best_cost = cost_val;
        selected_peak_idx = cand_idx;
      end
    end
  end
  F0 = selected_peak_idx;
  peaks_idx(r) = selected_peak_idx;
        l = selected_peak_idx;
  minI = l; maxI = l;
  while minI > 1 && spec_mclms(minI - 1) > 0
    minI = minI - 1;
  end
  while maxI < nfft && spec_mclms(maxI + 1) > 0
    maxI = maxI + 1;
  end
end
fmax = (nfft/2) / (nfft * beam.ipp_us * 1e-6 * beam.n_coh);
beam.M1 = rescale(peaks_idx, -fmax, fmax, InputMax=1024, InputMin=1);
beam.denoised_spectra = mclms_denoised_spectra;
beam.a_params.mccf_cost_scores = cost_scores;
beam.a_params.mccf_ST_W = ST_W_array;
beam.a_params.peaks_idx = peaks_idx;
```

```
new_beam = beam;
end
```

A.1.2 IME

```
% entry point
function beam = estimate_params(beam, cfg)
arguments (Input)
  beam RadarData
  cfg.vel_axis (1, :) double
end
arguments (Output)
  beam RadarData
end
if isempty(beam.noise_level)
  beam.noise_level = ime.HS_noise_estimate(beam.spectra, beam.in_coh);
end
candidates = ime.get_candidate_peaks(beam.denoised_spectra, beam.noise_level,
cfg.vel_axis);
delta_R = 3.0e8 * beam.m_fBaudLength_us * 1e-6 / 2;
profile = ime.trace_profile(candidates, beam.m_fOffZenith, delta_R);
beam.M1 = profile.velocity / (3e8 / 205e6 / 2);
beam.a_params.ime_candidates = candidates;
beam.a_params.ime_radial_vel = profile.velocity;
beam.a_params.ime_power = profile.power;
beam.a_params.ime_snr_db = profile.snr_db;
end

function noise_level = HS_noise_estimate(spectra, in_coh)
arguments(Input)
  spectra (:, 1024) double
  in_coh (1,1) double
end
arguments(Output)
  noise_level (1, :) double
end
[numRangeBins, N] = size(spectra);
noise_level = zeros(1, numRangeBins);
if in_coh < 1
  in_coh = 1;
end
for r = 1:numRangeBins
  A = sort(spectra(r, :), 'ascend');
  n = (1:N);
  Pn = cumsum(A) ./ n;
  secondMoment = cumsum(A.^2) ./ n;
  Qn = secondMoment - Pn.^2;
  Qn(Qn <= 0) = NaN;
  Rn = (Pn.^2) ./ (in_coh .* Qn);
  valid = find(Rn > 1);
  if isempty(valid)
    k = max(1, round(0.1*N));
  else
    k = max(valid);
  end
```

```
  noise_level(r) = Pn(k);
end
end

function candidates = get_candidate_peaks(Pclean, noise_level, vel_axis)
[numRangeBins, nfft] = size(Pclean);
candidates = cell(numRangeBins, 1);
total_bandwidth = max(vel_axis) - min(vel_axis);
half_width = (0.20 * total_bandwidth) / 2;
previous_velocity = NaN;
for r = 1:numRangeBins
  P = Pclean(r, :);
  Sn = noise_level(r);
  if r == 1
    [~, peak_idx] = max(P);
    center_velocity = vel_axis(peak_idx);
  else
    center_velocity = previous_velocity;
  end
  lower_velocity = center_velocity - half_width;
  upper_velocity = center_velocity + half_width;
  idx_window = find(vel_axis >= lower_velocity & vel_axis <= upper_velocity);
  if isempty(idx_window)
    candidates{r} = struct([]);
    continue;
  end
  Pworking = P(idx_window);
  selected = struct('idx', {}, 'velocity', {}, 'power', {}, 'snr_db', {},
'width', {});
  for p = 1:3
    if isempty(Pworking), break; end
    [max_amp, local_idx] = max(Pworking);
    if max_amp <= 0, break; end
    left = local_idx;
    while left > 1 && Pworking(left-1) > 0
      left = left - 1;
    end
    right = local_idx;
    while right < length(Pworking) && Pworking(right+1) > 0
      right = right + 1;
    end
    P_peak = Pworking(left:right);
    V_peak = vel_axis(idx_window(left:right));
    total_peak_power = sum(P_peak);
    mean_velocity = sum(P_peak .* V_peak) / total_peak_power;
    if Sn > 0
      total_noise_power = Sn * nfft;
      snr_linear = total_peak_power / total_noise_power;
    else
      snr_linear = Inf;
    end
    snr_db = 10 * log10(snr_linear);
    if snr_db <= 0, break; end
    newCandidate.idx = idx_window(local_idx);
```

```
    newCandidate.velocity = mean_velocity;
    newCandidate.power = total_peak_power;
    newCandidate.snr_db = snr_db;
    newCandidate.width = (right - left + 1);
    selected(end+1) = newCandidate;
    Pworking(left:right) = 0;
  end
  candidates{r} = selected;
  if ~isempty(selected)
    velocities = [selected.velocity];
    [~, closest] = min(abs(velocities - center_velocity));
    previous_velocity = velocities(closest);
  else
    if r == 1
      previous_velocity = center_velocity;
    end
  end
end
end

function profile = trace_profile(candidates, off_zenith, delta_R)
numRangeBins = length(candidates);
profile = struct('velocity', zeros(1, numRangeBins), ...
  'power', zeros(1, numRangeBins), ...
  'snr_db', zeros(1, numRangeBins));
w1 = 0.4;
w2 = 0.6;
group_size = 5;
delta_vm = max(0.5, 0.05 * sind(off_zenith) * delta_R);
for g = 1:ceil(numRangeBins / group_size)
  start_bin = (g - 1) * group_size + 1;
  end_bin = min(g * group_size, numRangeBins);
  bins_in_group = end_bin - start_bin + 1;
  cost_dp = cell(bins_in_group, 1);
  path_dp = cell(bins_in_group, 1);
  peaks_curr = candidates{start_bin};
  num_peaks = length(peaks_curr);
  if num_peaks == 0
    cost_dp{1} = 0;
    path_dp{1} = {1};
    peaks_curr = struct('velocity', NaN, 'power', NaN, 'snr_db', NaN);
    candidates{start_bin} = peaks_curr;
  else
    snr_max = max([peaks_curr.snr_db]);
    if snr_max <= 0, snr_max = 1; end
    cost_dp{1} = zeros(num_peaks, 1);
    path_dp{1} = num2cell(1:num_peaks)';
    for p = 1:num_peaks
      P_rj = peaks_curr(p).snr_db / snr_max;
      cost_dp{1}(p) = w2 * P_rj;
    end
  end
  for i = 2:bins_in_group
    r = start_bin + i - 1;
```

```
    peaks_prev = candidates{r-1};
    peaks_curr = candidates{r};
    if isempty(peaks_curr)
      peaks_curr = struct('velocity', NaN, 'power', NaN, 'snr_db', NaN);
      candidates{r} = peaks_curr;
    end
    num_prev = length(peaks_prev);
    num_curr = length(peaks_curr);
    snr_max = max([peaks_curr.snr_db]);
    if snr_max <= 0, snr_max = 1; end
    new_cost_dp = zeros(num_curr, 1);
    new_path_dp = cell(num_curr, 1);
    for c = 1:num_curr
      max_path_cost = -inf;
      best_prev_idx = 1;
      P_rj = peaks_curr(c).snr_db / snr_max;
      for p = 1:num_prev
        if isnan(peaks_curr(c).velocity) || isnan(peaks_prev(p).velocity)
          x = inf;
        else
          x = abs(peaks_curr(c).velocity - peaks_prev(p).velocity);
        end
        if x <= delta_vm
          T_x = 1;
        elseif x <= 2 * delta_vm
          T_x = (2 * delta_vm - x) / delta_vm;
        else
          T_x = 0;
        end
        transition_cost = w1 * T_x + w2 * P_rj;
        total_cost = cost_dp{i-1}(p) + transition_cost;
        if total_cost > max_path_cost
          max_path_cost = total_cost;
          best_prev_idx = p;
        end
      end
      new_cost_dp(c) = max_path_cost;
      new_path_dp{c} = [path_dp{i-1}{best_prev_idx}, c];
    end
    cost_dp{i} = new_cost_dp;
    path_dp{i} = new_path_dp;
  end
  [~, best_final_idx] = max(cost_dp{bins_in_group});
  best_path = path_dp{bins_in_group}{best_final_idx};
  for i = 1:bins_in_group
    r = start_bin + i - 1;
    selected_peak_idx = best_path(i);
    chosen_peak = candidates{r}(selected_peak_idx);
    profile.velocity(r) = chosen_peak.velocity;
    profile.power(r)    = chosen_peak.power;
    profile.snr_db(r)   = chosen_peak.snr_db;
  end
end
end
```

A.2 Presentation Slides

Implementation and Evaluation of Wind Profile Estimation Algorithms

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Project Overview

Project Guide: Dr Binu Paul
Project Phase: Phase 1
Research Area: Radar Signal Processing
Focus: Evaluation and comparison of MCCF, IME and hybrid method

Existing Problem

- Wind velocity estimation using the 205 MHz ST radar at CUSAT requires accurate identification of peaks in Doppler spectra.
- **Major challenges:**
  - Low signal-to-noise ratio (SNR)
  - Ground clutter and unwanted interference
  - Multiple spectral peaks and ambiguous peak selection
- Incorrect peak detection leads to errors in Doppler velocity estimation and discontinuities in wind profiles.
- **Objective:** Compare MCCF, IME, and hybrid methods to study their accuracy under different radar signal conditions.

Project Overview

Research Focus

Reliable Doppler spectral peak and moment estimation for the 205 MHz ST radar at CUSAT.

Methods Investigated

- **MCCF:** Mutual Convolution Cost Function
- **IME:** Improved Moments Estimation
- **Hybrid Methods:** Combining mutual convolution with IME processing (Hybrid 1 and Hybrid 2)

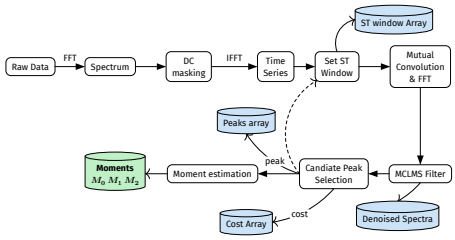
Main Objective

- Evaluate the methods on radar spectra.
- Evaluate performance using synthetic data with known parameters.

MCCF: Working Principle

- A Doppler spectral peak estimation method with noise suppression.
- Processing Stages
  - **Preprocessing:** FFT and DC-component removal.
  - **Noise Suppression:** Mutual convolution and MCLMS adaptive filtering.
  - **Peak Selection:** Candidate peaks are evaluated using a velocity-continuity and magnitude-based cost function.
  - **Moment Estimation:** Estimate the Doppler spectral moments from the selected peak.
- Improve spectral quality and select a suitable peak instead of relying only on the strongest spectral component.

MCCF: Algorithm Flow



MCCF: Important steps

- **Setting Space-Time window**: Limits peak search window to a selected Doppler range (space). (No time window is used due to the large time difference between observations)
- **Mutual Convolution**: Divides the single spectrum into slices of half length and convolves them together to generate multiple intermediate combined results. Pick the result with the highest peak
- **Modified Complex Least Mean Square Filter**: Isolate the stable, coherent features and remove random noise and ground clutter that fluctuate unpredictably across different delay steps

MCCF: Results and Observations

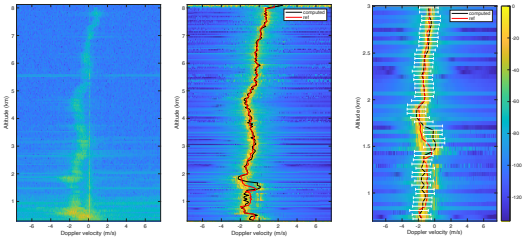


Figure 1: Original Spectrum Figure 2: Estimated Moments Figure 3: ST window

IME: Working Principle

A Doppler spectral estimation method designed to identify reliable atmospheric peaks under noisy and low-SNR conditions.

Main Processing Stages

- Spectral Preprocessing
  - Remove DC components and ground clutter.
  - Apply 5-point smoothing and estimate the noise level.
- Adaptive Doppler Window
  - Restrict peak searching to a selected Doppler range.
  - Identify prospective peaks using their SNR.
- Profile Tracing
  - Connect candidate peaks across adjacent range bins.
  - Select the profile using velocity continuity and relative SNR.

(b): Adaptive Doppler Window

Adaptive Window Selection

- Start from the lowest range bin.
- Use its maximum peak as the initial mean Doppler velocity.
- Centre the next Doppler window around the candidate closest to the previous mean velocity.

Prospective Peak Selection

- Detect peaks within the adaptive window
- Retain peaks with SNR > 0 dB (upto 3 Peaks).

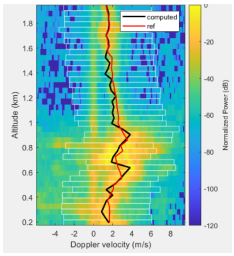


Figure 4: Adaptive Doppler window

(c): Grouping and Cost Allocation

1. Grouping and Link Construction
2. Cost Function: The link cost combines velocity continuity and normalised SNR:

$$C = w_1 T + w_2 P$$

3. Final Profile Selection
   - Select the trace with the maximum total cost.
   - Use its final velocity to initialise the next group.

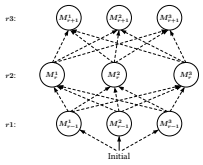


Figure 5: Grouping and profile tracing

IME: Results

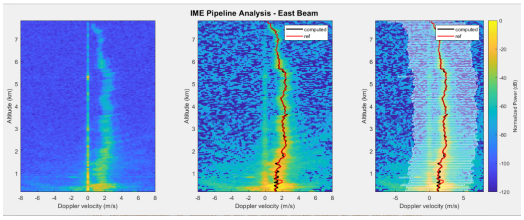


Figure 6: a) Original b) After applying IME c) Adaptive window

Hybrid Methods

Mutual convolution (MC) in MCCF provides noise suppression, while IME provides adaptive peak selection and profile tracing. The hybrid methods combine these processing stages.

Hybrid 1

MC + IME Denoising + IME Peak Selection

- Spectrum processing using MC
- IME denoising
- IME peak selection

Hybrid 2

MC + IME Peak Selection

- Spectrum processing using MC
- Direct IME peak selection

Synthetic Radar Data Generation

Generated from a predefined *UVW* wind-velocity array by reversing the conventional radar signal-processing pipeline to produce a raw data file.

Generate *UVW* wind velocities → Reverse radar processing pipeline → Synthetic raw radar file

- Provides known ground-truth wind velocities.
- Enables direct comparison of algorithm outputs against the predefined velocities.
- Allows controlled testing of algorithms without relying exclusively on newly acquired radar observations.

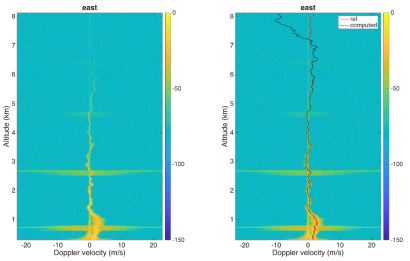


Figure 7: Synthesised raw data and corresponding velocity estimation using MCCF

Evaluvation with Synthetic data

Synthesised 500 raw data samples and applied it to 5 algorithms.

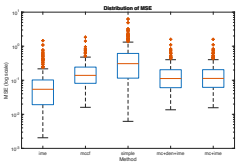


Figure 8: Mean Squared Error (MSE)

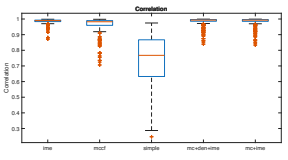


Figure 9: Correlation with ref data

Evaluvation with Synthetic data

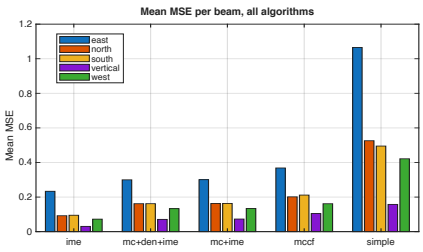


Figure 10: MSE in all beams

Evaluvation with Radiosonde

- Projection of radiosonde observation along with MCCF's estimation.
- In the winter season, the atmosphere is stable hence, the result closely matches

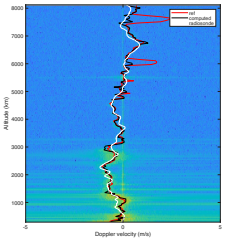


Figure 11: Observation on 20 Jan 2026, 14:20 IST

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Evaluvation with Data Pattern's output

Used 226 radar observations taken on 20 January 2026.

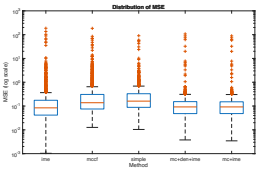


Figure 12: MSE

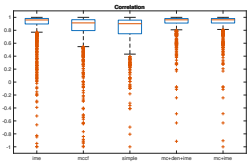


Figure 13: Correlation with reference data

Evaluvation with Data Pattern's output

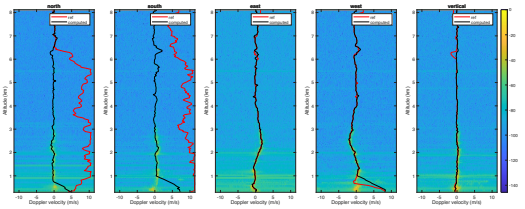


Figure 14: Case where Data Pattern's algorithm fell short, moments computed by MCCF

Conclusion

- Studied MCCF and IME for Doppler spectral peak and moment estimation using ST radar data.
- Examined two hybrid approaches combining mutual convolution with IME peak-selection and profile-tracing stages.
- The methods address challenges such as low SNR and unwanted spectral components.
- Methods were evaluated with various sources.

Future Scope

- Wavelet-based denoising: Improve weak Doppler peak detection under low-SNR conditions.
- ML-Based Peak Detection: Improve atmospheric echo identification in multi-peak spectra.
- Making synthesised radar data more realistic by adding more features
- Distinguish rain echoes from clear-air atmospheric echoes in Doppler spectra to improve the reliability of wind-velocity estimation
- Can be used to fill gaps between observations in the lower atmosphere while the experiment is being conducted at higher altitudes
- Use synthetic data to train reinforcement learning-based algorithms for estimating wind profiles.